

Reflexivity is unnecessary in Ricceri's compact-operator minimax identity

Candidate full-solution packet for arXiv:1508.04891, Remark 3

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Status: candidate full solution, likely valid, subject to human review. Theorem 2 of the source remains true on every infinite-dimensional Banach space. Thus the counterexample conjectured in a Schur space does not exist.

1 The question and the stronger answer

Ricceri's Theorem 2 assumes that E is an infinite-dimensional reflexive real Banach space and proves, for a nonzero compact linear operator $T : E \rightarrow E$, a finite-valued lower-semicontinuous convex coercive functional $\varphi : E \rightarrow \mathbb{R}$, a compact interval $I \ni 0$, and a finite-valued lower-semicontinuous convex $\psi : I \rightarrow \mathbb{R}$, that

$$\sup_{x \in X} \inf_{\lambda \in I} (\varphi(Tx - \lambda x) + \psi(\lambda)) = r + \psi(0), \quad X = \{x : \varphi(Tx) \leq r\}, \quad (1)$$

whenever $r > \varphi(0)$ [1]. Remark 3 asks whether reflexivity can be dropped and conjectures a counterexample in a Banach space with the Schur property.

The answer is stronger than requested.

Theorem 1 (Reflexivity-free identity). *Let E be any infinite-dimensional real Banach space, let $T : E \rightarrow E$ be a nonzero compact linear operator, and let $\varphi : E \rightarrow \mathbb{R}$ be finite-valued, lower semicontinuous, convex, and coercive. Let $I \subset \mathbb{R}$ be any compact set containing 0, and let $\psi : I \rightarrow \mathbb{R}$ be finite-valued and lower semicontinuous. Then (1) holds for every $r > \varphi(0)$.*

In particular, neither reflexivity of E , convexity of ψ , nor the interval structure of I is needed. The conclusion therefore holds on ℓ_1 and on every other infinite-dimensional Schur space.

2 Two elementary lemmas

Lemma 1 (Compact approximate kernel). *If $H \subset E$ is an infinite-dimensional closed subspace and $T : E \rightarrow E$ is compact, then for every positive sequence $\varepsilon_n \downarrow 0$ there are $z_n \in H$ with $\|z_n\| = 1$ and $\|Tz_n\| \leq \varepsilon_n$.*

Proof. Otherwise $T|_H$ is bounded below by some positive constant. Its inverse on $T(H)$ is then Lipschitz, while compactness makes the image of the unit ball of H relatively compact. The unit ball of H would be relatively compact, contradicting infinite-dimensionality. $\square$

Lemma 2 (Intermediate range level). *For every $s > \varphi(0)$ there is $y \in E$ such that $\varphi(Ty) = s$.*

Proof. Choose $v \neq 0$ in $T(E)$. A finite-valued lower-semicontinuous convex functional on a Banach space is norm-continuous. Hence $t \mapsto \varphi(tv)$ is continuous, equals $\varphi(0)$ at zero, and tends to $+\infty$ as $t \rightarrow +\infty$ by coercivity. The intermediate value theorem gives $t \geq 0$ with $\varphi(tv) = s$; write $tv = Ty$. $\square$

3 Proof of the theorem

Proof of Theorem 1. The upper bound is immediate: for $x \in X$, the admissible choice $\lambda = 0$ gives

$$\inf_{\lambda \in I} (\varphi(Tx - \lambda x) + \psi(\lambda)) \leq \varphi(Tx) + \psi(0) \leq r + \psi(0).$$

Fix an arbitrary s with $\varphi(0) < s < r$. By Lemma 2, choose $y \in E$ and put

$$a = Ty, \quad \varphi(a) = s.$$

Continuity and convexity of φ give a subgradient $p \in E^*$ at a :

$$\varphi(w) \geq \varphi(a) + p(w - a) \quad (w \in E). \quad (2)$$

The closed subspace $H = \ker p$ is infinite-dimensional (also when $p = 0$). Apply Lemma 1 in H to choose

$$z_n \in H, \quad \|z_n\| = 1, \quad \|Tz_n\| \leq n^{-2}.$$

Set $x_n = y + nz_n$. Then

$$Tx_n = a + nTz_n \rightarrow a,$$

so $\varphi(Tx_n) \rightarrow s < r$. After discarding finitely many indices, $x_n \in X$.

For each such n , lower semicontinuity on compact I gives a minimizer $\lambda_n \in I$ and

$$m_n := \min_{\lambda \in I} (\varphi(Tx_n - \lambda x_n) + \psi(\lambda)) = \varphi(Tx_n - \lambda_n x_n) + \psi(\lambda_n).$$

We claim

$$\liminf_{n \rightarrow \infty} m_n \geq s + \psi(0). \quad (3)$$

It suffices to start with an arbitrary subsequence and pass to a further one on which $\lambda_n \rightarrow \lambda \in I$.

If $\lambda \neq 0$, then Tx_n is bounded, while $\|x_n\| \geq n - \|y\|$; hence $\|Tx_n - \lambda_n x_n\| \rightarrow \infty$. Coercivity makes the φ term tend to $+\infty$. Since a finite-valued lower-semicontinuous function on compact I is bounded below, this case gives more than (3).

Suppose now that $\lambda = 0$ and put $c_n = n\lambda_n$. If (c_n) is unbounded, pass further so that $|c_n| \rightarrow \infty$. From

$$Tx_n - \lambda_n x_n = a + nTz_n - \lambda_n y - c_n z_n$$

and $n\|Tz_n\| \leq n^{-1}$, the residual norm again tends to infinity, so coercivity applies.

It remains only the case where (c_n) is bounded. Since $p(z_n) = 0$,

$$p(Tx_n - \lambda_n x_n - a) = n p(Tz_n) - \lambda_n p(y) \rightarrow 0.$$

The supporting inequality (2) therefore gives

$$\liminf_n \varphi(Tx_n - \lambda_n x_n) \geq \varphi(a) = s.$$

Also $\liminf_n \psi(\lambda_n) \geq \psi(0)$ by lower semicontinuity at zero. This proves (3) in every possible subsequential regime.

Consequently,

$$\sup_{x \in X} \inf_{\lambda \in I} (\varphi(Tx - \lambda x) + \psi(\lambda)) \geq \liminf_n m_n \geq s + \psi(0).$$

Letting $s \uparrow r$ gives the lower bound $r + \psi(0)$, which combines with the first paragraph to prove equality. $\square$

4 Why the construction works

There is one delicate parameter scale. If a minimizing λ_n stays away from zero, then the term $-\lambda_n n z_n$ escapes to infinity. The same is true if $\lambda_n \rightarrow 0$ but $n|\lambda_n| \rightarrow \infty$. Only $\lambda_n = O(1/n)$ can keep the residual bounded. At exactly that scale, the supporting functional detects all first-order displacement except the large z_n term, because z_n was chosen in $\ker p$. Compactness makes the remaining error nTz_n vanish.

This also explains why reflexive weak compactness in the source proof is not intrinsic to the identity. It is one route to a general minimax theorem, but the particular compact-operator expression admits a direct near-kernel construction.

5 Novelty and limitations

On 11 August 2026, exact searches of the run registry and cheap indexes found no prior packet or known answer. Searches across the locally parsed arXiv corpus found only the source and later bibliographic citations. A bounded citation-index query returned two citing works. Inspection of the primary texts—Ricceri’s 2020 gradient-systems preprint [2] and *Miscellaneous Applications of Certain Minimax Theorems II* [3]—found no discussion or resolution of Remark 3. This is a bounded novelty check, not an exhaustive priority claim.

The theorem retains the source assumptions that T is nonzero and compact, E is infinite-dimensional, and φ is finite-valued, convex, and coercive. The source already notes that infinite-dimensionality and compactness cannot simply be removed. The present result addresses only the reflexivity question in Remark 3; it does not alter the general minimax Theorem 1.

A Source-page evidence

Figure 1 reproduces the source page containing Theorem 2 and Remark 3.

References

- [1] B. Ricceri, *A minimax theorem in infinite-dimensional topological vector spaces*, arXiv:1508.04891v3 [math.FA], 2015; Linear Nonlinear Anal. 2 (2016), 47–52.
- [2] B. Ricceri, *An alternative theorem for gradient systems*, arXiv:2002.01413v2 [math.AP], 2021.
- [3] B. Ricceri, *Miscellaneous Applications of Certain Minimax Theorems II*, arXiv:1810.08957 [math.FA], 2018; Acta Math. Vietnam. 45 (2020), 515–524.

where

$$X = \{x \in E : \varphi(T(x)) \leq r\} .$$

PROOF. By a classical result, φ turns out to be continuous with respect to the strong topology of E . Since $T(E)$ is a non-zero linear subspace, the set $\varphi(T(E))$ is unbounded above. Indeed, if not, φ would be constant on $T(E)$, contrary to the coercivity of φ . As a consequence, since $T(E)$ is connected, we have

$$\varphi(T(E)) = \left(\inf_{T(E)} \varphi, +\infty \right[.$$

From this, we clearly infer that

$$\sup_{x \in X} \varphi(T(x)) = r . \quad (2)$$

Next, consider the function $f : X \times I \rightarrow \mathbf{R}$ defined by

$$f(x, \lambda) = \varphi(T(x) - \lambda x) + \psi(\lambda)$$

for all $(x, \lambda) \in X \times I$. Now, denote by τ the weak topology of E . Notice that T , being linear and compact, turns out to be sequentially continuous from E with the topology τ to E with the strong topology. It is easy to check that this is equivalent to the continuity of T from E with the topology τ_s to E with the strong topology. Of course, (E, τ_s) is a Hausdorff topological vector space. Now, we are going to apply Theorem 1 to the function f considering E with the topology τ_s . First, notice that the set X is convex and its interior in τ_s is non-empty. Actually, X contains the non-empty set $T^{-1}(\varphi^{-1}(]-\infty, r[))$ which is open in τ_s , by the remarks above. Next, observe that, for each $\lambda \in \mathbf{R}$, the function $x \rightarrow T(x) - \lambda x$, being continuous and linear, is continuous from the weak to the weak topology, and so, *a fortiori*, from the τ_s to the weak topology. Of course, this implies that the function $(x, \lambda) \rightarrow T(x) - \lambda x$ is continuous from the product of τ_s and the topology of $\mathbf{R}$ to the weak topology. But then, since φ is weakly lower semicontinuous, the function f is lower semicontinuous in $X \times I$ with respect to the considered topology. Of course, f is convex in I . Finally, by a classical result, the spectrum of T is countable, and so the set, say D , of all $\lambda \in I$ such that $x \rightarrow T(x) - \lambda x$ is a homeomorphism between E (with the strong topology) and itself is dense in I . Fix $\lambda \in D$. Of course, since φ is coercive, for each $\rho \in \mathbf{R}$, the set

$$\{x \in E : \varphi(T(x) - \lambda x) \leq \rho\}$$

is bounded. Hence, due to the reflexivity of E , the sub-level sets of $f(\cdot, \lambda)$ are weakly compact and so, by the Eberlein-Smulyan theorem, sequentially weakly compact which is equivalent to sequentially τ_s -compact. Therefore, each assumption of Theorem 1 is satisfied and hence we have

$$\sup_{x \in X} \inf_{\lambda \in I} (\varphi(T(x) - \lambda x) + \psi(\lambda)) = \inf_{\lambda \in I} \sup_{x \in X} (\varphi(T(x) - \lambda x) + \psi(\lambda)) . \quad (3)$$

Now, observe that if $\lambda \in I \setminus \{0\}$, we have

$$\sup_{x \in X} \varphi(T(x) - \lambda x) = +\infty . \quad (4)$$

Indeed, since the τ_s -interior of X is non-empty and E is reflexive and infinite-dimensional, X turns out to be unbounded. But $T(X)$ is bounded (since φ is coercive) and so, since $\lambda \neq 0$,

$$\sup_{x \in X} \|T(x) - \lambda x\| = +\infty$$

which yields (4) by the coercivity of φ again. At this point, the conclusion follows directly from (2), (3) and (4). $\triangle$

REMARK 2. - Notice that both infinite-dimensionality of E and compactness of T cannot be dropped in Theorem 2. In this connection, it is enough to take $T(x) = x$, $I = [0, 1]$, $\varphi(x) = \|x\|$, and $\psi = 0$.

REMARK 3. - At present, we do not know any example showing that the reflexivity of E cannot be dropped. However, we conjecture that such an example can be constructed in infinite-dimensional Banach spaces with the Schur property.

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Figure 1: Source paper: Theorem 2 and Remark 3, including the conjectured Schur-space counterexample.